



Review

Greenhouse gases emissions and global climate change: Examining the influence of CO₂, CH₄, and N₂O

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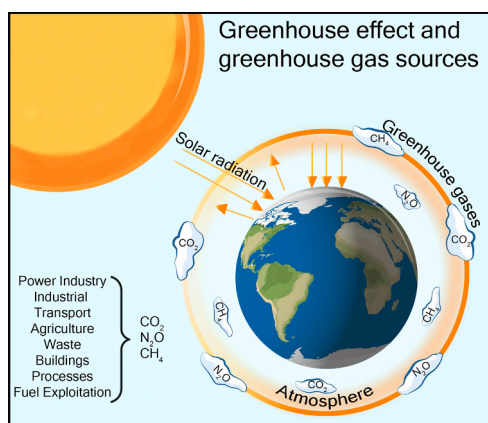
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HIGHLIGHTS

- A review of the major factors in GHG emissions.
- An overview of the major determinants of GHG emissions.
- Summarizes important research on global climate change.
- Provides an historic perspective of global warming in the context of the industrial revolution.
- Integrates carbon sequestration in the analysis of global climate change.

GRAPHICAL ABSTRACT



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ABSTRACT

An in-depth analysis of the role of greenhouse gases (GHGs) in climate change is examined here along with their diverse sources, including the combustion of fossil fuels, agriculture, and industrial processes. Key GHG components such as carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) are considered, along with data on emissions across various economic sectors. The consequences of climate change are also highlighted, ranging from more frequent and intense extreme weather events to rising sea levels and impacts on ecosystems and human health. The industrial revolution and unrestricted use of fossil fuels are key factors leading to an increase in GHG concentrations in the atmosphere. Global efforts to reduce emissions are considered, starting with the 1997 Kyoto Protocol and culminating in the 2015 Paris Agreement. The limited effectiveness of early initiatives is underscored, emphasizing the significant importance of the Paris Agreement that provides a global framework for establishing goals to reduce GHG emissions by country. The Green Climate Fund and other international financial mechanisms are also considered as essential tools for financing sustainable projects in developing

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countries. The global community faces the challenge and necessity for more ambitious efforts to achieve the set goals for reducing GHG emissions. Successful strategies are examined by Sweden, Costa Rica, and Denmark to achieve zero GHG emissions that integrate renewable energy sources and technologies. The importance of global cooperation for creating a sustainable future is also emphasized.

1. Introduction

The greenhouse effect is a phenomenon occurring in the Earth's atmosphere under the influence of solar radiation. The sun emits energy, including visible light, ultraviolet rays, and infrared radiation, that penetrates the atmosphere mainly composed of nitrogen, oxygen, water vapor, and various gases, including greenhouse gases (GHGs). Interaction with infrared radiation plays a crucial role in creating the greenhouse effect (Mitchell, 1989; Rodhe, 1990; Ramanathan and Vogelmann, 1997; Smith et al., 2003; Besson et al., 2010; Zhong and Haigh, 2013; Anderson et al., 2016; Raghuraman et al., 2021; Wee et al., 2024).

The essence of the mechanism is as follows: solar radiation, mainly in the form of visible light, passes through the atmosphere and reaches the Earth's surface. There, the energy from the sun is absorbed by land, water, and vegetation, and re-emitted as heat in the form of longer-wave infrared radiation (Kaltenegger, 2017). Normally, this radiation would escape into space. GHGs, such as carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and others, are capable of absorbing a portion of the infrared radiation emitted by the Earth (Ledley et al., 1999). These pollutants, which can last in the atmosphere for years to centuries, trap the heat. The greenhouse effect is a direct consequence of this heat no longer radiating into space.

The interaction of infrared radiation with GHG molecules, primarily CO₂, CH₄, and N₂O, induces vibrational or rotational motions in these molecules. This internal motion imparts energy to the molecules that is radiated back into the atmosphere (Mitchell, 1989). This process generates additional thermal radiation, increasing the temperature of the surrounding air and the Earth's surface (Ming et al., 2014). The greenhouse effect is similar to how glass in a greenhouse allows visible light to pass through but traps thermal radiation, leading to an increase in temperature inside the greenhouse.

Characteristics of the Earth's surface upon which sunlight is directed plays a crucial role in the atmospheric greenhouse effect. Forests, meadows, oceans, ice, deserts, and cities all emit radiation with varying intensity. The snow-white surface of a glacier reflects a significant portion of solar radiation back into space, thereby minimizing surface heating on the ice. In contrast, dark soil absorbs solar radiation intensively, contributing to significant warming of the surface and emitted radiation. Additionally, cloud cover influences the greenhouse effect by reducing both the amount of solar radiation reaching the Earth's surface and the amount of radiant energy emitted into space.

The increase in GHG concentrations as a result of human activity enhances the greenhouse effect, leading to global warming. This process influences the climate, causing changes in weather patterns, sea levels, and other aspects of the environment. Combating climate change involves reducing GHG emissions and exploring sustainable solutions for the future.

2. Significance and main objectives of the review

This review holds particular significance in light of global efforts to combat climate change. The aim of this article is to conduct a comprehensive analysis of GHG emissions, addressing various aspects such as emission sources, their environmental impact, and measures taken to mitigate them. The review covers a wide range of topics, including CO₂, CH₄, N₂O, that contribute significantly to global warming.

The article examines the main sectors responsible for GHG emissions, including energy, industrial production, transportation, agriculture,

waste management, and construction. The authors analyze the complex mechanisms of emission generation in each sector and study the various types of GHGs that contribute to overall climate change.

Also explored is the impact of GHGs on global ecosystems, biodiversity, and human health. This study incorporates data from recent research concerning rising levels of CO₂ and CH₄ in the atmosphere and explores possible solutions. An important aspect of the review is the examination of natural carbon sinks, such as forests, and their role in mitigating the effects of GHG emissions.

Additionally, the review includes an assessment of international initiatives such as the Paris Agreement, as well as national policies and strategies aimed at reducing GHG emissions. The authors also discuss new technologies and innovations that can contribute to sustainable development.

The article is aimed at a broad range of readers, including scientists, environmental and climate specialists, as well as students and early-career researchers who want to gain a general understanding of the issues related to GHG emissions. For professionals in the field, the review can be a source of inspiration and ideas for further research and development. For the general public, the article offers a clear and comprehensive explanation of the complexity of GHG emission issues and possible solutions.

Thus, this article should be useful for anyone interested in climate change issues, offering a comprehensive perspective on the problem, including scientific, technical, and political aspects.

3. Methodology

To ensure a comprehensive understanding of the role of GHGs in climate change and the related global efforts to reduce emissions, a systematic approach was adopted in the literature review process. The study utilized various databases and search engines, including PubMed, Google Scholar, Scopus, and Web of Science, to gather a wide range of sources such as academic journals, government and international reports, books, and conference materials. To reflect current research in this field, the search focused on literature published in the past 20 years, though key foundational works from earlier periods were also included. Keywords used for the search included "greenhouse gases", "global warming", "climate change", "carbon dioxide", "methane", "nitrous oxide", "GHG emissions", "Kyoto Protocol", "Paris Agreement", and "climate change mitigation".

Articles were selected based on topic relevance, publication quality, and data freshness. The review included works that examined the connection between GHG emissions and climate change, emission data from various economic sectors, as well as international climate agreements. Additionally, studies exploring the role of forests and other natural systems in carbon sequestration were analyzed.

Studies not directly related to the main topics of GHG emissions and climate change, or non-peer reviewed sources, were excluded. Also excluded were articles focused solely on local environmental issues or with limited relevance to the broader climate change impacts.

Data on emission levels, climate trends, and other relevant information were gathered from reliable sources such as the Intergovernmental Panel on Climate Change (IPCC), the Emissions Database for Global Atmospheric Research (EDGAR) and the United States Environmental Protection Agency (EPA).

Data collected from various sources were analyzed to identify trends, key factors, and significant findings related to GHG emissions, the impact of climate change, and mitigation measures. This analysis

included evaluating the contribution of different industries, changes in emission levels over time, and the effectiveness of international climate agreements.

By adopting this comprehensive methodology, the paper offers a thorough examination of the impact of GHG on climate change, providing insights into emission sources, their environmental impact, and global efforts to mitigate these effects.

4. Global efforts to reduce greenhouse gas emissions

International efforts to reduce GHG emissions have involved several significant initiatives and agreements. The Kyoto Protocol, adopted in 1997, imposed commitments on developed countries to reduce GHG emissions. However, due to its limited scope and non-participation by developing countries, this approach proved to be ineffective (Lau et al., 2012). In response to these limitations, the Paris Agreement was reached in 2015, providing a comprehensive framework for participating countries to set national GHG reduction targets. It acknowledged the special responsibility of developed countries and provided financial support for developing nations.

Under the Paris Agreement, countries agreed to take measures to limit global warming to “well below” 2 °C above pre-industrial levels (Rogelj et al., 2016). Considering the potential impact on climate and ecosystems, there was also a commitment to strive for efforts to limit warming to 1.5 °C. These temperature goals underscore the seriousness of commitments to reducing GHG emissions and each country's contribution to global efforts to combat climate change. However, the current concentration of GHG is projected to lead to temperature increases by the end of this century significantly above the targets set by the Paris Agreement, accompanied by more extreme weather conditions.

Additionally, in this context, international financial mechanisms play a crucial role, such as the Green Climate Fund for developing countries, established to finance projects aimed at reducing GHG emissions and adapting to climate change (Hitz and Smith, 2004; Van Kerckhoff et al., 2011; Adger et al., 2013; Antimiani et al., 2017; Dean et al., 2018; Busch et al., 2022; Chien et al., 2023). Furthermore, regional and bilateral agreements complement these efforts, providing more targeted and adapted emission reduction strategies, as well as facilitating the exchange of experiences and technologies among countries.

These agreements, therefore, serve as important instruments in the global effort to mitigate the impact of GHG and ensure a sustainable future for the planet. Efforts to reduce GHG emissions involve diverse strategies and measures at the national government and global community levels, aimed at transitioning to more sustainable methods of energy production and reducing the adverse effects on the Earth's climate system.

International initiatives to reduce GHG emissions find practical embodiment in the strategies of various countries that successfully implement innovative approaches to sustainable energy. One example of a successful case is Sweden that actively integrates renewable sources such as wind and hydroenergy, and incentivises the purchase of electric vehicles, challenging the necessity of relying on fossil fuels (Zhong et al., 2021a, 2021b). Costa Rica, fully embracing renewable energy sources, including hydroelectric power, solar, and wind installations, serves as a shining example of a country that has fully transitioned to clean energy (Godínez-Zamora et al., 2020). Denmark, focusing efforts on wind energy and innovation, has achieved impressive results in creating a sustainable energy system (Korberg et al., 2020). Key features of its approach include significant investments in wind energy, the development of energy-efficient technologies, and the stimulation of energy consumption from renewable sources.

These countries serve as vivid examples of successful strategies in reducing emissions in line with the commitments outlined in the Paris Agreement. Their experiences underscore the importance of global cooperation, investment in sustainable technologies, and the commitment to the shared goal of creating a sustainable future for our planet.

These successful initiatives provide undeniable evidence that transitioning to sustainable methods of production and consumption benefits not only the environment but also the economy and society as a whole.

Global efforts to reduce GHG emissions require a better understanding of the problem and collaboration and knowledge exchange among countries and organizations. This path not only creates favourable conditions for future generations but also contributes to economic growth, improves public health, and ensures global sustainability (Nordahl et al., 2020). The responsibility for our future lies in global efforts aimed at preserving our planet and ensuring a sustainable existence for all.

5. Primary greenhouse gases

GHG, such as CO₂, CH₄, and N₂O, are capable of capturing and retaining thermal radiation in the Earth's atmosphere. They play a primary role in the greenhouse effect. CO₂, CH₄, and N₂O retain heat due to their ability to absorb and emit thermal infrared radiation. This process is associated with the specific molecular structure of these gases.

CO₂ has a linear arrangement between a single carbon and two oxygen atoms. This dipole structure facilitates an interaction with infrared radiation (Galashev, 2011; Kirk-Davidoff, 2018). CO₂ also has a long atmospheric lifetime of 50 to 200 years (Inman, 2008). It absorbs and emits thermal radiation in specific wavelength ranges (Harries et al., 2008). When thermal infrared radiation enters the atmosphere, CO₂ molecules absorb this energy and re-emit it in various directions, including back to the Earth's surface. This process contributes to the retention of heat in the atmosphere.

N₂O, composed of two nitrogen (N) atoms and one oxygen (O) atom, also has a linear structure. A relatively long-lived gas at about 114 years (IPCC, 2001), it can also emit infrared radiation. While its contribution to the greenhouse effect is not as significant as either CO₂ or CH₄ (Tian et al., 2011), it plays a crucial role in atmospheric chemistry and affects the climate through aerosol particle formation, influencing cloud properties, chemical reactions, and the atmospheric heat balance. Additionally, N₂O impacts ozone levels and regulates methane concentrations that also influences climate (Ravishankara et al., 2009). The climate impact of N₂O is complex and dependent on various factors, including atmospheric conditions and geographical location.

In contrast to the linear structures of CO₂ and N₂O, CH₄ consists of one carbon (C) atom and four hydrogen (H) atoms in a tetrahedral arrangement. It is a more efficient greenhouse gas compared to CO₂ (Jeffrey et al., 2021) and has a shorter atmospheric lifetime of only about 12 years (EPA, 2023). Its molecular structure allows it to both absorb and emit infrared radiation, effectively trapping heat and making it a significant component of the greenhouse effect.

Only about 45 % of CO₂ that is emitted remains in the atmosphere (Raupach et al., 2014). The remainder is absorbed: 25 % by the ocean (Watson et al., 2020), and approximately 30 % by terrestrial ecosystems such as forests (Duffy et al., 2021). This balance can vary from year-to-year due to natural climate cycles and other factors (Peters et al., 2017).

Taking into account the heat-trapping characteristics measured in terms of global warming potential, CH₄ is 23 times more effective, and N₂O is 296 times more effective than CO₂ (IPCC, 2001; Liu and Greaver, 2009). However, the amount of CO₂ in the Earth's atmosphere significantly exceeds the amounts of CH₄ or N₂O making it of much greater concern. The continued accumulation of CO₂ in the atmosphere makes it the most troubling of the three major GHGs. The problem will exacerbate with continued emissions. Considering the long lifespan of CO₂ as a GHG, the achieved temperature level will persist for several decades even if emissions were reduced to zero.

While these three GHGs play a crucial role in maintaining the Earth's warmth, an excess of these gases, as a result of human activities, leads to the intensification of the greenhouse effect, climate change, and global warming.

6. Greenhouse gas sources

The primary source of CO₂ is the combustion of fossil fuels such as coal, oil, and natural gas. Compounding the effect of CO₂ emissions, deforestation is a significant factor in carbon dioxide absorption as trees can sequester a significant amount of CO₂ (Koerner and Klopatek, 2002; Raupach et al., 2007; Ho et al., 2011; Raihan and Tuspekova, 2022). Anthropogenic sources of CH₄ include activities related to the extraction and transportation of natural gas, organic waste processing in landfills, and livestock farming (Ramachandra et al., 2015; Gong and Shi, 2021; Shahzad et al., 2021). N₂O emissions occur in agriculture, industrial processes, and fuel combustion (Smith, 2017; Tian et al., 2020; Jovani-Sancho et al., 2023), with fertilizers used in agriculture being a major contributor to these emissions. Water vapor and ozone also contribute, regulated by natural processes and anthropogenic activities.

Human activities significantly increase the concentration of GHG in the atmosphere, intensifying the greenhouse effect and impacting the climate (Althor et al., 2016; Zheng et al., 2019). According to data from Table 1, provided by the Emission Database for Global Atmospheric Research (EDGAR) and developed by the European Commission's Joint Research Centre (Crippa et al., 2018), the main sources of these emissions are concentrated in several key sectors, each contributing differently to the overall GHG gas emissions. The power industry is a leading source, mainly from the burning of fossil fuels such as coal, oil, and natural gas for electricity generation (Lamb et al., 2021; Liu et al., 2023). Industrial combustion in sectors like chemicals, steel, and cement also increases GHG emissions, with CO₂ resulting from fuel combustion and chemical reactions, along with CH₄ and N₂O from various processes

Table 1
Main sources of GHG emissions in various sectors and types of emissions.

Sector	Major GHG sources	Types of GHG emissions
Power Industry	Power plants: burning coal, oil, gas for electricity generation	CO ₂ (from combustion), CH ₄ (from gas extraction and transportation), N ₂ O (from combustion processes)
Industrial Combustion	Chemical industry: synthesis of chemicals; Steel and cement production	CO ₂ (from limestone and clay calcination), CH ₄ (from fertilizer and plastic production), N ₂ O (from synthesis processes)
Transport	Road transport, aviation, maritime transport: burning gasoline, diesel, kerosene	CO ₂ (from fuel combustion), N ₂ O (especially from aviation kerosene combustion), CH ₄ (from diesel and gas combustion)
Agriculture	Agriculture, livestock farming: rice fields, livestock farms	CO ₂ (from agricultural practices), CH ₄ (from anaerobic fermentation in rice fields and enteric fermentation in livestock), N ₂ O (from fertilizer use and soil processes)
Waste	Waste disposal, garbage incineration, wastewater treatment Decomposition of organic waste in landfills.	CO ₂ (from organic material decomposition), CH ₄ (from anaerobic fermentation in landfills and waste incineration)
Buildings	Energy consumption, Heating	CO ₂ (from electricity usage), CH ₄ (from heating systems)
Processes	Cement production, Iron and steel industry	CO ₂ (from cement production and steel manufacturing), CH ₄ (from construction waste decomposition and demolition), N ₂ O (from material production processes)
Fuel Exploitation	Oil and gas extraction, mining operations	CO ₂ (from disruption of geological formations), CH ₄ (from the extraction and transportation of hydrocarbons)

(Hong et al., 2021).

Transportation contributes to emissions through the combustion of gasoline, diesel, and kerosene in cars, airplanes, and ships, releasing CO₂, N₂O, and CH₄. Agriculture generates GHG from rice cultivation, livestock farming, and fertilizer use, with significant emissions of CH₄ and N₂O (Nayak et al., 2015; Kim et al., 2016; Menegat et al., 2022). The waste sector produces GHG through the decomposition of organic waste and its incineration, while buildings emit CO₂ due to energy use and CH₄ from natural gas-based heating systems (Zhong et al., 2021a, 2021b). Industrial processes and fuel exploitation, including oil and gas extraction, add to these emissions (Malik et al., 2016; Grasso, 2019). These diverse sources emphasize the need to implement strategies to reduce GHG emissions to combat climate change and promote sustainable development. Understanding the contributions from each sector, as illustrated in Table 1 provided by EDGAR, is crucial for developing effective mitigation measures to address this global challenge.

Aquatic ecosystems, especially in tropical regions, play an important role in the emission of GHGs, such as CO₂ and CH₄ (Dalal and Allen, 2008; Teodoru et al., 2015; Kumar et al., 2020; Hui et al., 2022; Deng et al., 2023). These emissions result from natural processes like organic matter decomposition and anaerobic fermentation, as well as from anthropogenic impacts. In tropical zones, where temperature and biological activity are higher, the decomposition of organic matter is accelerated, leading to increased CO₂ emissions.

Wetlands and other water-rich ecosystems common in the tropics release CH₄ due to methanogenesis, which occurs under low oxygen conditions. Furthermore, human factors like deforestation, land use changes, and runoff from agricultural and industrial waste can also boost GHG emissions from aquatic ecosystems (Ran et al., 2021). For instance, nutrient pollution from agricultural runoff can lead to eutrophication, which can increase CO₂ and CH₄ emissions (Bhattacharyya et al., 2022; Chan et al., 2024). Thus, the impact of aquatic ecosystems on global GHG emissions is significant, especially in tropical regions, and should be taken into account when developing strategies to combat climate change.

Fig. 1 represents GHG emissions in gigatons of carbon dioxide equivalents (GtCO₂eq), reflecting the contributions of various sectors to overall pollution levels. EDGAR data shows that the power industry is the single greatest contributor with 14.67 GtCO₂eq or 38.1 % of overall emissions. Transport, including car emissions primarily from larger vehicles, at 7.97 GtCO₂eq or 20.7 % of the total, also plays a crucial role in climate change. Industrial combustion and processes contribute 6.54 (17 %) and 3.23 GtCO₂eq (8.4 %), respectively, emphasizing the

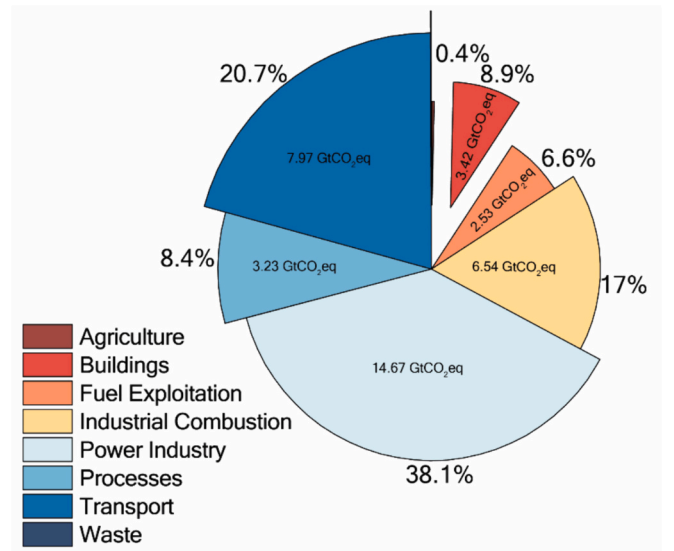


Fig. 1. Contribution of greenhouse gas emissions by sector.

significance of industrial operations. The contribution of buildings, with 3.42 GtCO₂eq or 8.9 %, points to the significant impact of construction and building operations. Agriculture emits 0.15 GtCO₂eq or 0.39 % of total emissions, a very limited contribution to overall emissions. The waste processing sector has the smallest contribution of 0.02 GtCO₂eq (0.04 %). The total emissions in 2022 amounted to 38.52 GtCO₂eq, underscoring the need for collective efforts and decisive measures across various sectors to mitigate climate change.

From the early 1970s to 2022, there has been a consistent trend of increasing emissions of CO₂, CH₄, and N₂O (see Fig. 2). The primary gas responsible for climate change, constituting 73.5 % of the total emissions is CO₂, with emissions rising with a pronounced increase after 2000. Accounting for 21.5 % and 4.9 %, respectively, CH₄ and N₂O emissions are also on the rise but at a lower rate. Despite a temporary reduction in CO₂ emissions in 2020 due to the COVID-19 pandemic (Liu et al., 2020; Le Quéré et al., 2020; Ray et al., 2022), there is a new surge observed in 2021 and 2022. These data underscore the urgent need for effective measures to reduce GHG emissions to mitigate the impact of anthropogenic activities on the climate and environment.

The increase in GHG emissions after 2000 can be explained by rapid economic growth and industrialization in developing countries, leading to increased industrial production and energy consumption (Waheed et al., 2019; Sarkodie et al., 2020). Increased transportation activities and the use of inefficient methods of transport have also contributed to emissions growth (Lamb et al., 2021). Additionally, agriculture and changes in land use have contributed to increases in CH₄ and N₂O emissions. Global economic and social factors, along with insufficient regulatory measures and the slow transition to sustainable energy sources, also play a significant role in this trend (Liobikienė and Butkus, 2019). Addressing this issue requires concerted efforts and global strategies to reduce dependence on fossil resources and transition to sustainable practices and technologies.

7. Forests as a key factor in mitigating climate change

The rise in GHG is not only a result of increased emissions, it is also caused by less CO₂ being sequestered by plants. Forests, covering approximately 30 % of the Earth's surface, play a key role in mitigating climate change, as trees absorb carbon from the air during their growth (Trumbore et al., 2015). A recent study (Harris et al., 2021) revealed that forests absorb approximately 7.6 GtCO₂eq, which is nearly 1.5 times the annual CO₂ emissions of the United States (4.8 GtCO₂eq) and four times more than Russia (1.9 GtCO₂eq). Tropical forests, mainly located in South America (Amazon), Africa (Congo Basin), and South-east Asia, absorb more CO₂ than boreal forests in Canada and Siberia (Schimel et al., 2015; Artaxo et al., 2022; Chen et al., 2024). However, it's worth noting that this figure may vary from year-to-year and can be subject to changes due to various factors, including land use, forest fires,

climate change, etc.

Tropical forests surpass boreal forests in carbon sequestration due to high plant biodiversity, rapid plant growth, high biomass density, and soil microorganism activity (Lejeune et al., 2018; Tagesson et al., 2020). The warm and humid tropical climate provides constant conditions for intensive plant growth and efficient photosynthetic activity (Lewis et al., 2009; Osland et al., 2021). High biomass density and soil microorganism activity contribute to maximum carbon absorption and retention in the ecosystem, making tropical forests particularly significant in the context of climate change mitigation (Arnell et al., 2016; Pecl et al., 2017; Mitchard, 2018; Mora et al., 2018; Cook-Patton et al., 2020; Baldrian et al., 2023). In contrast, boreal forests are characterized by lower temperatures and slower carbon turnover with only a 3 to 4 month growing season, reducing the ability to absorb CO₂. Therefore, preserving tropical forests in their pristine state can contribute to additional temperature reduction within the framework of global warming.

Unlike other sectors where CO₂ is emitted into the atmosphere in one direction, forests act as a two-way transport flow. They absorb CO₂ during their growth or regrowth and release it when cut down or degraded. However, human-induced deforestation over the last 8000 years has led to significant carbon emissions. This process can be offset through afforestation (planting forests in new areas), reforestation (planting trees in cleared areas), and natural forest regeneration, which has great potential for retaining carbon in plants and soil. Combining these approaches can be key to successfully combating climate change.

8. Conclusions

Several important conclusions can be made from a comprehensive review of GHG, the sources of emissions, carbon sequestration, climate impact, and the on-going global effort to reduce emissions.

First and foremost, it is clear that of GHGs such as CO₂, CH₄, and N₂O are closely tied to climate change. The primary sources of these gases include the burning of fossil fuels, industrial processes, and to a much lesser extent, agriculture. Deforestation also plays a role by reducing the amount of carbon sequestered in vegetation. Emphasis must be placed on adopting a systematic approach to emissions reduction, covering a wide range of measures across different sectors of the economy.

The consequences of climate change pose serious challenges to the global community. Extreme weather conditions, rising sea levels, ecosystem stress, threats to agriculture, and human health all demand urgent measures to mitigate their effects.

International efforts to reduce emissions, particularly the Kyoto Protocol and the Paris Agreement, play a crucial role in forming a collective front against climate change. However, there is a need for more effective strategies and coordinated actions, especially considering that current emissions of GHGs exceeds established targets.

Financial support for developing countries provided by international mechanisms such as the Green Climate Fund plays a key role in achieving global emission reduction goals. Nevertheless, the focus is on the need for additional resources and strategies to achieve ambitious goals, such as keeping global warming “significantly below” 2 °C compared to pre-industrial levels.

This review underscores that successful emission reduction strategies are already being implemented in several countries, such as Sweden, Costa Rica, and Denmark. These positive examples demonstrate that a transition to sustainable methods of production and consumption is possible and brings benefits not only to the environment but also to the economy and society as a whole.

Further collective efforts are needed on a global scale to create a sustainable future. Global initiatives, international agreements, effective strategies, and the exchange of experiences – all these elements are an integral part of the response to the challenge of climate change. The universal responsibility for preserving the planet lies in the power of unity, coordination, and the adoption of balanced measures towards a sustainable future.

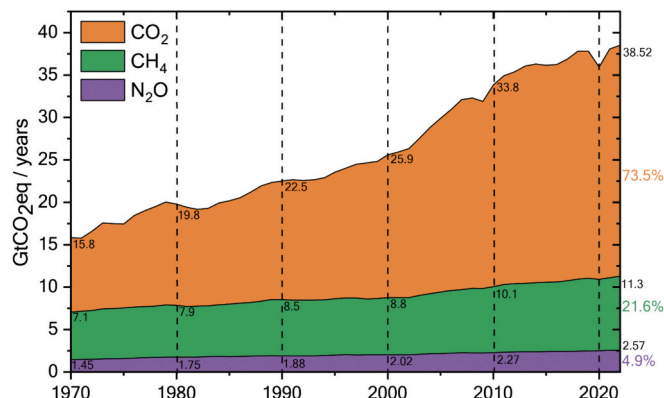


Fig. 2. Changes in GHG emissions from 1970 to 2022 obtained to EGRID.

CRediT authorship contribution statement

Mikalai Filonchyk: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Michael P. Peterson:** Writing – review & editing, Writing – original draft, Visualization, Validation, Conceptualization. **Lifeng Zhang:** Visualization, Validation, Methodology, Investigation. **Volha Hurynovich:** Data curation, Writing – review & editing. **Yi He:** Writing – review & editing, Writing – original draft, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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